

EDUCATION

BTech - Data Science, Business and Economics	Plaksha University, Mohali	9.12	2024-Present
12th - CISCE	Sri Sri Academy, Kolkata	95%	2024
10th - CISCE	Sri Sri Academy, Kolkata	97.8%	2022

EXPERIENCE

- Risk Management Intern | Motilal Oswal

May 2026 - July 2026

 - Built an **Early Warning System** across **170 NBFCs** in **5 lending verticals** to fix reactive monitoring, enabling proactive risk detection.
 - Partnered with traders to automate derivatives analysis via a Python-based NSE Option Workbench, cutting analysis from **2.5 to 1 hour**.
 - Enabled faster spread construction using **Black 76** pricing within the Workbench, supporting **stress testing** across **20 scenarios**.
- Data Analyst Trainee | Buyogo Technologies India Pvt Ltd

May 2025 - July 2025

 - Built **6 Metabase dashboards** with **25+ KPIs** using factory-, mechanic-, and date-level filters to track maintenance and inventory.
 - Built **SQL-based** inventory analyses by integrating maintenance ticketing and stock data to improve **inventory visibility**, enabling efficient stock-keeping and procurement decisions.
 - Designed a **mechanic ranking system** to benchmark performance, flag bottlenecks, and drive productivity gains across factories.

PROJECTS

- CityConscience: AI Municipal Reporting Platform | Prof. Anupam Sobti & Dr. Siddharth

April 2026-Present

 - Built an AI-powered civic platform to fix Mohali's lack of infrastructure prioritization, validated via a **27-user pilot** across **15 zones**.
 - Engineered a **multimodal AI workflow** (Gemini 2.5 Flash) to fix slow complaint handling, automating routing across **14 departments**.
 - Layered in gamification and GenAI features to fix low citizen participation and drive **sustained engagement** across the platform.
- Mutual Fund Volatility Prediction using Financial News Sentiment | Dr. Siddharth

May 2026

 - Developed a quantitative **volatility forecasting framework** to enhance risk management using news sentiment data across **19 equities**.
 - Built an ML pipeline using **17 engineered features** and rigorous leakage controls to ensure robust, reliable predictions.
 - Benchmarked the XGBoost pipeline against **RiskMetrics EWMA**, delivering **10.89% higher forecasting accuracy** for risk decisions.
- COLE Optimizer Research Project | Dr.Ashish Kumar Pandey

Dec 2025

 - Fixed slow, **unstable convergence** in neural network training, a bottleneck **driving up compute cost**, by building a novel optimizer.
 - Engineered **rolling-window plateau detection** with **patience-based learning-rate decay** to stabilize convergence and reduce iterations.
 - Benchmarked against Adam and RMSprop, achieving **95.9% accuracy** with greater consistency than standard optimization methods.

POSITIONS OF RESPONSIBILITY

- Vice President | Entrepreneurship Cell@ Plaksha University

August 2025 - Present

 - Planned Plaksha's **first-ever** national-level **entrepreneurship summit, Aeternum '25**, overseeing sponsorships and execution for a **400+ participant turnout**.
 - Managed stakeholder coordination for **Trifecta 2026**, Plaksha's **business conclave**, aligning operations and communication across three club leaderships while managing **external brand relationships**.
 - Led brand partnership talks with **Crowwd** and **Finlatics**, securing key E-Cell collaborations.
- Member | Plaksha Consulting Club

August 2024 - Present

 - Represented the Consulting Club in **8+ intercollegiate case competitions**, applying structured frameworks to solve real-world business challenges.
 - Organized case competitions, casebook sessions, and case interview workshops to build practical consulting skills among club members.
 - Drove the **highest attendance recorded** for a club event, strengthening **consulting culture** and engagement across campus.

ACHIEVEMENTS

- Dean's List (2024–25)** – Ranked in the **top 5%** of the cohort based on **cumulative GPA**.
- Ranked 5th globally** and won a **\$2,500 prize** in the **Global Scaling Challenge**, organized by the University of New Mexico, competing against **60+ teams** from **25+ countries** by developing a go-to-market, pricing, and supply chain strategy for a deep-tech startup.
- 1st Place Winner**, Startup Showdown 2026 –**national-level entrepreneurship case competition** hosted by Ashoka University, competing against **50+ participants**.
- 1st Runner-Up**, Plaksha Inter-College Basketball Tournament (2025); represented the university in several intercollegiate competitions.

SKILLS

Financial Analysis: Financial Statement Analysis, Credit Risk Analysis, Black-Scholes Option Pricing, Fixed Income Analysis. | **Programming:** Python (Pandas, NumPy), SQL | **Data & BI:** Microsoft Excel, Metabase